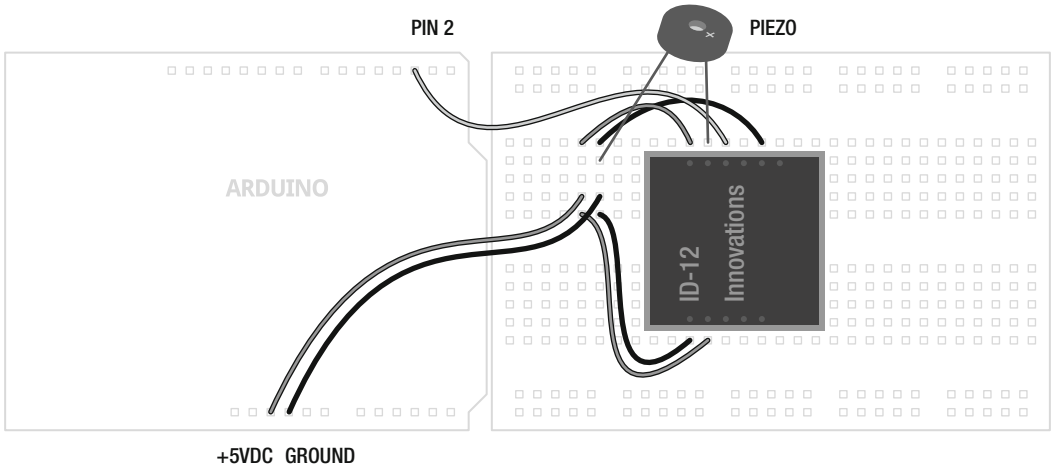


Figure 10-4. *RFID Reader illustration*

Uploading the Source Code

The source code in Listing 10-2 builds on the basic principals of serial communications that we have been discussing so far, but adds some new wrinkles. First, the reader sends a 16-byte stream of data in the form of hexadecimal ASCII characters each time a tag has been read, but we only really want the unique 12-digit hexadecimal ID number. Second, because this data is sent one byte at a time, we need to read each byte individually and piece it together into a single character string. Finally, we might want to know the tag's ID, compare different tags to known ID numbers, and do certain things based on the tag number. For example, maybe we want to let one cat inside but not the other, or we want our fridge to tell us when we drank the last beer. Following is our source code for this project and a discussion of how it works and the functions that we've used.

Listing 10-2. *RFID Reader Source Code*

```
#include <SoftwareSerial.h>

SoftwareSerial rfid(2,3);

char tag01[] = "4500B8F08489";
char tag02[] = "4500B8D36947";

char tagString[13];

void setup() {
  Serial.begin(9600);
  rfid.begin(9600);
}
```